

- speaker tuning headroom,
- latency in certain workflows.

Sensor hubs: “always-on” sensing without killing battery

A sensor hub (often called a context hub) is a low-power processor that processes sensor data so the main application processor can sleep, saving power.

This affects:

- step counting and fitness tracking,
- raise-to-wake, tap gestures,
- crash detection / motion-triggered features,
- “why my battery drains overnight” mysteries.

This is why: Premium-feeling phones often feel premium because background intelligence is efficient – sensor hubs and low-power processing aren’t glamorous, but they’re quality-of-life kings.

TV/streaming chips: why “smart TV performance” varies wildly

A smart TV is basically a purpose-built computer optimised for decoding video, processing images, and running a UI. TV SoCs typically include:

- CPU/GPU for the OS and UI,
- video decoder/encoder blocks,
- memory bandwidth management,
- picture-quality engines for scaling/de-interlacing and more.

Manufacturers like MediaTek openly position smart TV SoCs around features like high-refresh support, AI picture processing, and integrated TV-specific blocks.

And on the more affordable end, vendors like Amlogic are widely used for TV and media player SoCs, with configura-

tions that integrate CPU/GPU, memory support, and hardware video decoding.

Here’s what you need to know:

- If the UI lags, it’s often CPU + memory constraints.
- If playback struggles, it’s often codec support and decoder blocks.
- If motion looks odd, it can be picture pipeline and processing.

Step-by-step: quick “TV chip sanity” checks in a showroom

1. Open the TV’s app store and scroll fast.
2. Open settings menus and switch between tiles rapidly.
3. Start a 4K YouTube HDR demo (if available) and observe: dropped frames, audio sync.
4. Switch between apps quickly (YouTube → streaming app → home).

If it stutters here, it won’t magically improve at home.

Vehicle/EV electronics: motor control, BMS, and safety silicon basics

Modern vehicles – especially EVs – are rolling electronics platforms. Two chip-heavy subsystems matter immediately:

Battery Management System (BMS)

A BMS monitors and controls the state of a battery pack to keep it operating safely and effectively over time.

BMS functions typically include monitoring, protection, balancing, and thermal management.

Power electronics (inverters, DC-DC stages)

EVs and charging infrastructure rely on efficient switching devices (power semiconductors). Silicon carbide (SiC) power MOSFETs are used in EV traction inverters, on-board chargers, and DC-DC converters, helping efficiency in these power stages.

India context: As EV adoption rises, the “chip content” of vehicles becomes a massive value lever – especially for range, charging behaviour, reliability, and cost.

Home power + inverter silicon: chargers, UPS/inverters, solar – decoded

Indian homes interact with power electronics more than many markets: inverters/UPS systems, solar inverters, voltage stabilisers, fast chargers, and power banks.

Power semiconductors (like MOSFETs and modules) matter because they switch power efficiently at high voltage/current. For example, SiC MOSFETs are positioned as suitable for high-power, high-efficiency applications like solar inverters and UPS.

Pro tip: When an inverter runs cooler, lasts longer, or wastes less power, that’s not “brand magic” – it’s often power stage design choices and semiconductor selection. 

